

Executive Summary – Customer Churn Analysis

This analysis was done to understand **why customers are leaving** and what factors influence their decision to churn. After cleaning and exploring the data, it was found that **around 26–27% of customers have churned**, which means **nearly one out of every four customers stops using the service**. This clearly shows that customer retention needs attention.

Demographic & Tenure Insights

When we look at customer behavior, **tenure stands out as one of the strongest factors**. Customers who are new to the service are far more likely to churn, while customers who have stayed longer tend to remain loyal. Churn drops steadily as tenure increases, showing that **once customers are engaged for a longer period, they are less likely to leave**. Senior citizens also show a **slightly higher churn tendency**, suggesting that this group may need more support or simpler service experiences.

Service Usage & Churn Behavior

Service-related features play a very important role in retention. Customers who **do not have OnlineSecurity, TechSupport, OnlineBackup, or DeviceProtection** show **much higher churn percentages** compared to those who use these services. On the other hand, customers who subscribe to these support and security services are **more likely to stay**, which indicates that these features add real value and trust.

Entertainment & Connectivity Services

Internet service type also shows clear differences. **Fiber optic users account for the highest share of churn**, even though it is a premium service. In comparison, **DSL users churn less**, suggesting that factors such as pricing, expectations, or service quality may be impacting fiber customers more. Streaming services like **StreamingTV and StreamingMovies show a smaller impact on churn**, meaning entertainment add-ons alone do not strongly influence whether a customer stays or leaves.

Key Business Takeaways

- **~26–27% churn rate** signals an urgent need for retention-focused strategies
- **Low-tenure and senior customers** are the most vulnerable segments
- **Support and security services reduce churn significantly** and should be promoted early
- **Fiber optic churn is comparatively high**, warranting deeper pricing or service-quality analysis
- **Service bundling improves retention**, supporting cross-sell and upsell initiatives

Conclusion

Overall, the charts clearly show that **customer experience and service support matter more than basic demographics**. Focusing on **early customer engagement, promoting support services, and improving the experience for fiber optic users** can help reduce churn significantly and improve long-term customer retention